

Prelab2 Report

Table No. _____ Name & Stu No. _____ Score _____ of 100

PART ONE Light-Emitting Diode (LED) Characteristics (I-V Analysis)

LEDs are made from exotic semiconductor compounds. The actual semiconductor compound used in forming the PN junction during the manufacturing process determine the wavelength so as to determine the color of a LED. LEDs are available in a wide range of colors, the most common being green, yellow, orange, or red light. The LED emits visible light when forward biased. LEDs switch off and on rapidly, are very rugged and efficient, have a very long lifetime, don't heat up, and are easy to use. LEDs are widely used as visual indicators, displays, and light transmitters. Fig.1 shows the LED electronic symbol, lead identification and external package.

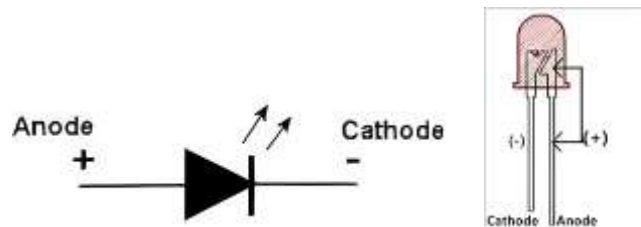


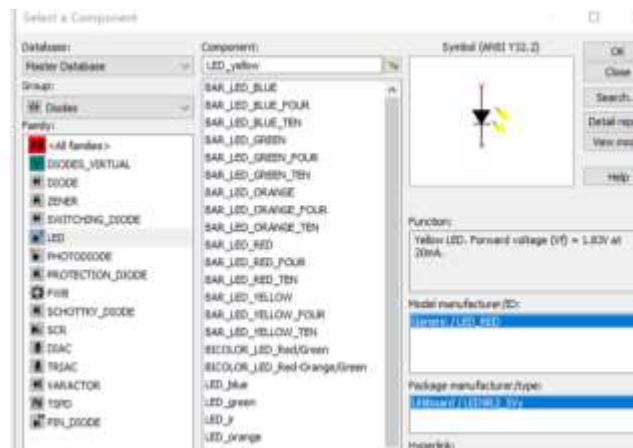
Figure 1. LED electronic symbol, lead identification and external package

NI Multisim simulation: I-V curve and Threshold Voltage

1. The LED is basically a specialized type of PN junction diode so they have very similar electrical characteristics to a PN junction diode. This means that an LED will pass current in its forward direction but block the flow of current in the reverse direction.

Now simulate the I-V characteristics of the **LED_red** which will be used in this lab.

- (1) Open Multisim and select the **LED_red** device from the “Place Diode” button.



- (2) Next, select the IV-Analysis tool from the “Instruments” toolbar (the same toolbar where you find the DMM).

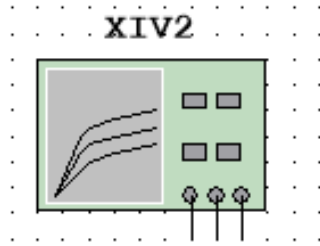


Fig. 2 (a) IV-Analysis tool

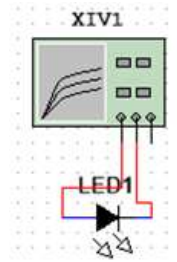


Fig. 2 (b) Multisim schematic for diode I-V curve

(3) Wire the device and instrument as below in Fig. 2(b).

(4) Double-click the IV-Analysis tool and set the “Simulate parameters” as shown in Fig. 2.

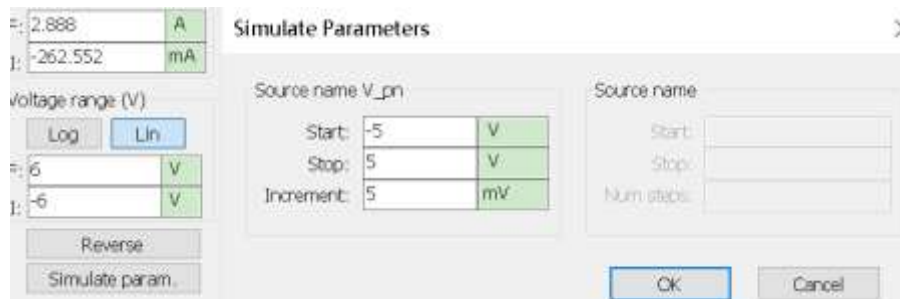
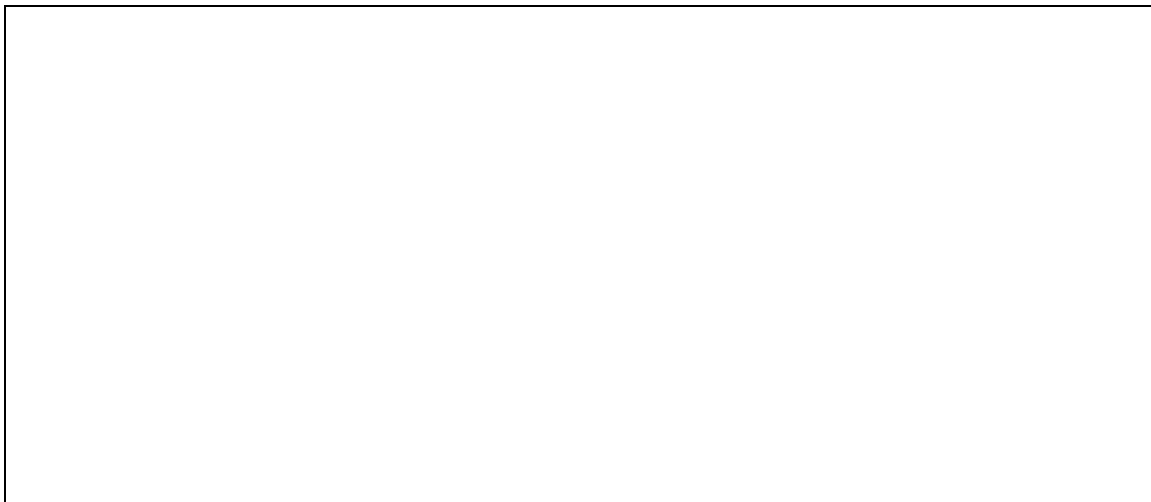


Fig. 3 Simulate parameters

(5) Simulate the circuit and draw the I-V curve. (6pts)



(6) Use your I-V curve to determine the threshold voltage of the LED_red.

Simulated Threshold Voltage Value: _____ (4pts)

PART TWO Correct LED Connection

Reverse Bias

When the voltage is applied in the reverse direction, that is anode (+) connected to the negative terminal and cathode (-) connected to the positive terminal, the diode blocks the current flow, acting as an open switch. This is known as the reverse bias.

Forward Bias

When the voltage across a diode is applied in the forward direction, that is anode (+) connected to the positive terminal and cathode (-) connected to the negative terminal, it allows current to flow easily, similar to a closed switch. This is known as the forward bias. **But! Never connect an LED directly to a battery or power supply!** When an LED is forward biased to the threshold of conduction, its current increases rapidly. It will be destroyed almost instantly because too much current will pass through and burn it out. LEDs must have a resistor in series to limit the current to a safe value. The light brightness is quite linearly proportional to the forward LED current.

The simplest circuit to drive an LED to light is through a series resistor as shown in Fig.4. The circuit must provide sufficient current to light the LED at the required brightness, but at the same time, it must limit the current to prevent damaging the LED.

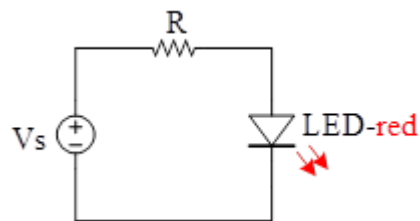


Fig.4 Simple LED (Light Emitting Diode) circuit diagram

LEDs have a maximum continuous current rating I_{max} . This is often around 25 or 30 mA. Running an LED near its rated maximum current gives the maximum brightness, at the cost of power dissipation heat. If you want to lower the power, you can usually just pick a current that is only one tenth of the rated maximum current.

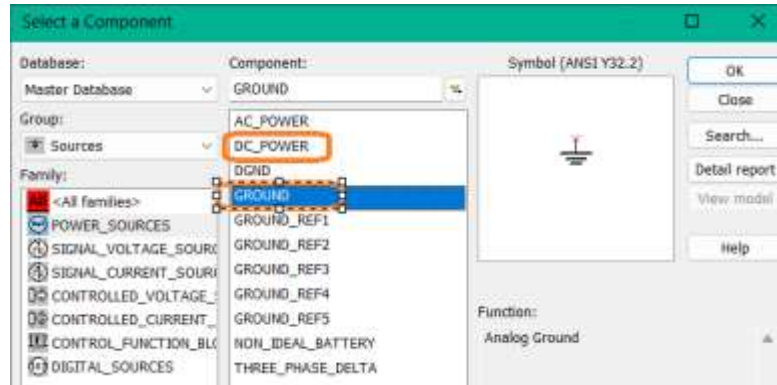
The voltage drop across the LED is approximately constant over a wide range of operating current. LEDs have a characteristic called “forward voltage” V_f . This forward voltage is the amount of voltage “lost” in the LED when operated at a certain reference current, usually defined to be about 20 mA. V_f depends primarily on the color of the LED, red, orange and yellow LEDs have a V_f of about 1.8 V.

1. Use Ohm's law and Kirchhoff's circuit laws to calculate the appropriate resistor value to aim for the current through LED_red is 20 mA—slightly under the maximum current. Light the LED with a $V_s=6.8V$, what resistor value we need? (4pts)

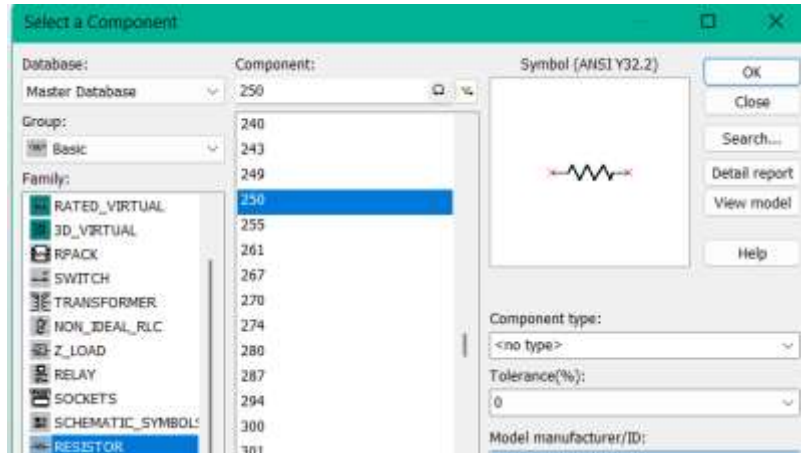
2. Build the circuit above in Multisim, $V_s=6.8V$, the resistance of R is the result in step1.

(1) Place > Components, you will see a component selection window that pops up where you can place your components and sources. In the Select Component windows, under Group:

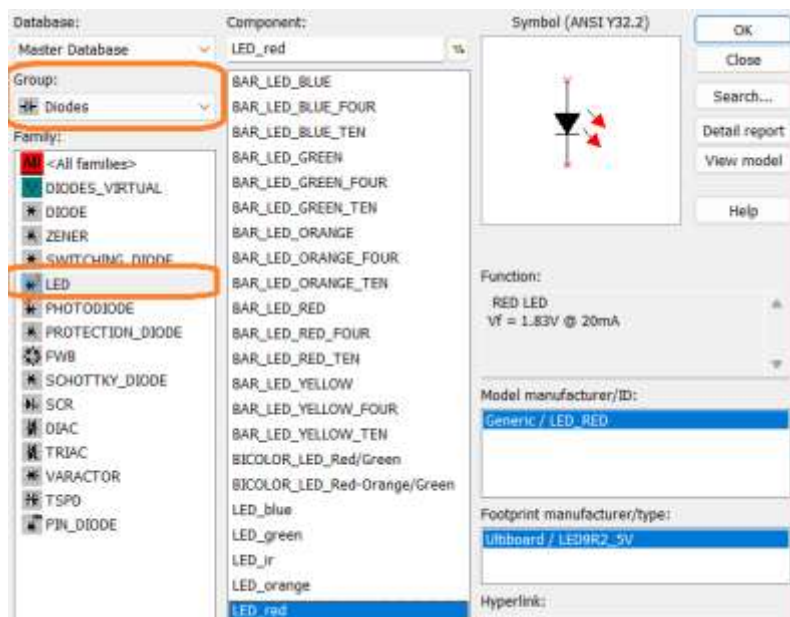
Select Sources for DC_POWER and GROUND



Select Basic for RESISTOR

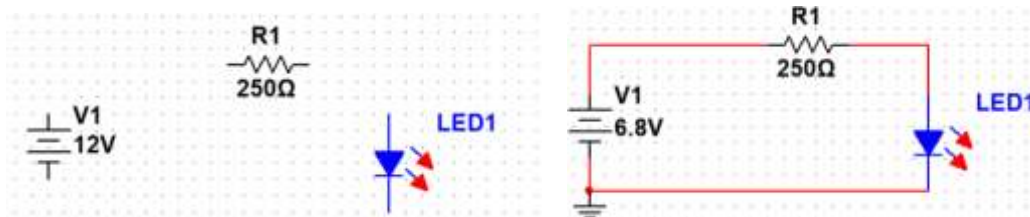


Select Diode for LED red



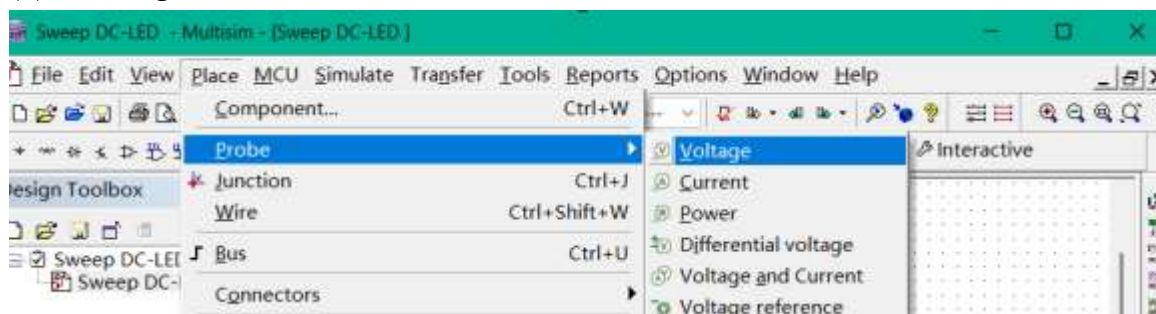
(2) Changing Component Parameters.

To change the values or names of the component, you can either right click on the component and select properties, or double click on the component, and a screen will pop up allowing you to change the properties and values of the component.

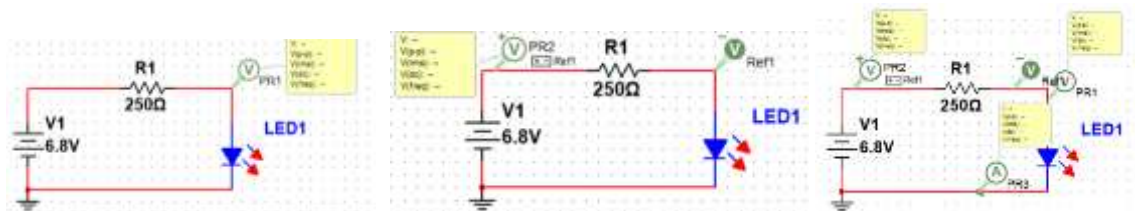


(3) Connect the wires to complete the circuit. To create a wire, place cursor at the node on an element and you should see a black dot appearing, click the mouse once and drag your wire to the edge of another component where a red dot should appear, click again to connect.

(4) Adding Measurements. Place > Probe, and choose.



a) Probe “Voltage ” defaults to measuring the voltage at that node relative to ground. So place a probe Voltage PR1 to measure the V_f (V_{LED}), since the bottom of LED is the ground node which do not need a voltage reference node here.



b) But, if we want to measure the voltage across R1, that is we actually want to measure the voltage relative to the bottom of that resistor. Now Select Probe “Differential Voltage ”, firstly place on one side of the resistor, then at the other side of R1.

(5) Next, add a current probe to the node between the LED and the Ground.

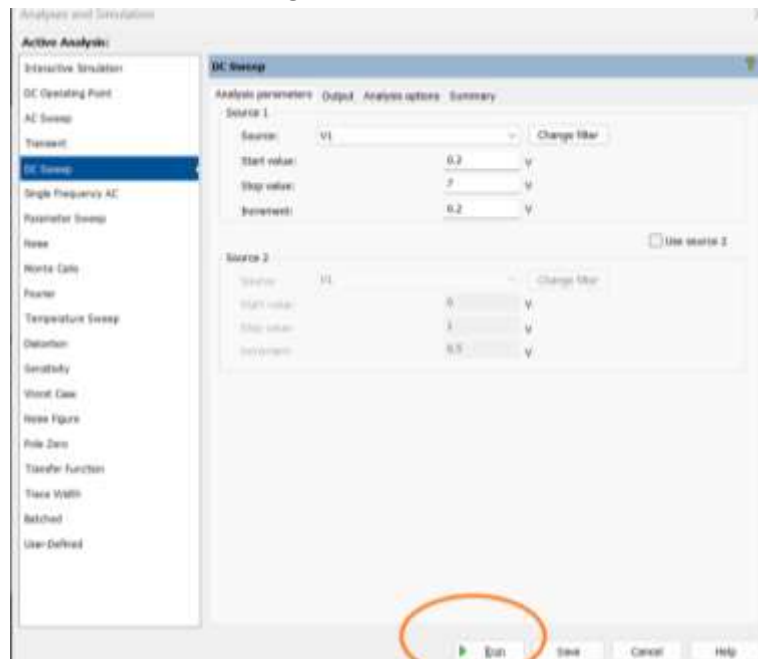
(6) Saving. To save your project, go to- File > Save as > Name your design

(7) Run simulation, record the results. (6pts)

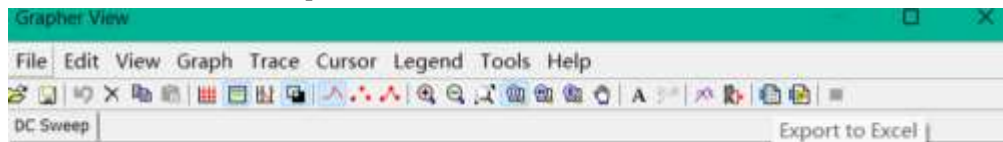
The voltage across R1	V_f	current I

(8) Parametric Sweep: To see the different Current I value and V_f values when we change the V_1 values, that is we can do a parametric sweep, go to- Simulate > Analyses and Simulation>

DC Sweep. In the Parameter Sweep window, setting as following Figure. After setting the parameters in the Analysis parameters, clicking on the Run, the Grapher View should pop up showing the Graph with 3 different curves. Click on the show Cursors will allow you to see the different current values, different voltage values for different V1 values.



(9) Click the Excel icon to export the data.



(10) Using the data obtained above, complete the table. (30pts)

V_i/V	V_f/V	V_R/V	I/mA	V_i/V	V_f/V	V_R/V	I/mA
0.20				1.90			
0.40				2.00			
0.60				2.50			
0.80				3.00			
1.00				4.00			
1.20				5.00			
1.40				6.00			
1.60				6.80			
1.70				6.90			
1.80				7.00			

(11) Please provide a commentary on your findings. (6pts)

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PART Three Correct LED Connection

(1) Design circuits to light two LED_reds in two methods. (10pts)

Series Circuits	Parallel Circuits

(2) Compare the two circuits. What are the advantages and disadvantages? Which one is better? (6pts)

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PART FOUR AC signals

- Time period: the time taken for the signal to complete one cycle. It is measured in seconds (s), but time periods tend to be short so milliseconds (ms) and microseconds (μ s) are often used.
- Frequency: inverse of the period of the signal, the number of cycles per second; units are cycles per second (Hz), but frequencies tend to be high so kilohertz (kHz) and megahertz (MHz) are often used.
- Amplitude: Amplitude is a non-negative scalar measure of a wave's maximum magnitude of oscillation. In electrical engineering it may be thought of as the maximum absolute value reached by a waveform as measured from the center of the oscillation.

- **Peak voltage:** Peak is the height of an AC waveform as measured from the center of the oscillation to the highest positive or lowest negative point on a graph.
- **Vpp:** Peak to peak Voltage value of the signal, twice the peak voltage (amplitude). It is measured in volts.
- **DC Offset:** Constant voltage added to the signal to increase or decrease its mean or average level. It is measured in volts.
- **RMS:** Root Mean Square, and is a way of expressing an AC quantity of voltage or current in terms functionally equivalent to DC.

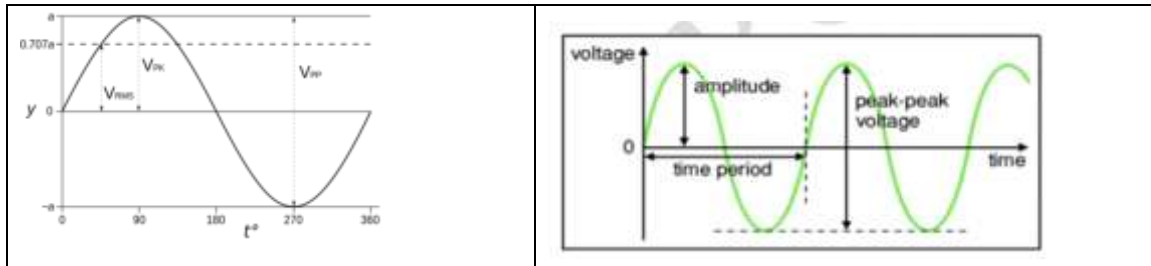
Some description about Sine Wave can be found on Chapter 9 of Fundamentals of Electric Circuits. A sinusoidal voltage source produces a voltage that varies as a sine wave with time, which can be represented mathematically as: $v = V_m \sin(\omega t + \theta)$

$\omega = 2\pi f$ is the radian frequency [radians/sec]

f is the waveform frequency in cycles per second [Hz]

V_m is the amplitude, peak voltage peak value [volts],

θ is the relative phase shift [radians] with respect to where you define time zero.

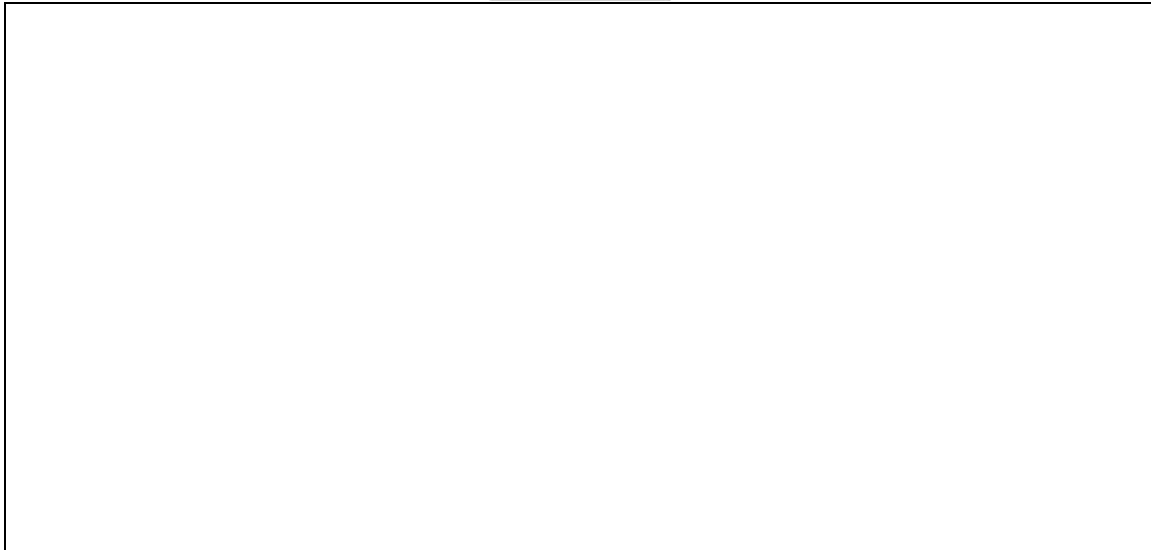
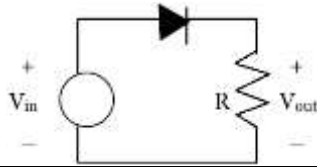


For a sinusoidal voltage $v(t) = 1.5 \times \sin(2\pi \times 1000 \times t)$ V, what is the frequency? what is the period? What is the amplitude? What is the peak Voltage? What is the RMS Voltage? What is the peak-peak voltage? Show your work. (6pts)

PART FIVE Half-Wave Rectifier

1. Look at the circuit below. Suppose V_{in} is a sine wave that has amplitude voltage 1.5V, frequency $f=1\text{kHz}$, and offset = 0V. Suppose the threshold voltage of the diode is $V_{\text{threshold}} =$

550mV. Describe what you will see at the output. Draw the Waveform of V_{in} and V_{out} on the same coordinate axis. Make sure to include the amplitude, frequency, and offset in terms of the input and V_{th} . (10pts)



PART SIX Capacitor Characteristics

There are two types of capacitors, electrolyte and non – electrolyte capacitors. The difference in the use of the two types of capacitors is that non-electrolytic capacitors can be charged in any direction, whereas the Electrolytic ones can only be charged in one direction. Electrolytic Capacitors are Polar; i.e., one of its two plates is Positive and other is Negative, whereas in non-electrolytic capacitors, both the plates are same, having no polarity.

1. Electrolytic capacitor, read the capacitance directly and record.
2. Non – electrolyte capacitors: use NUMERICAL CODES to specify their capacitance. Usually, these codes are 3 digit long, specifying the capacitance in Pico Farads; the first two digits are Tens and Units, whereas the third digit is power of 10.



For example, a code 102 means: $10 \times 10^2 = 1000 \text{pF} = 1 \text{nF}$ (not 102pF!)

1. Read the capacitance value below. (4pts)



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Questions

1. A description of the possible applications of the LED. (4pts)

2. What is the color of an LED is determined by? Is by the coloring of the 'package' (the plastic body). (4pts)